Supplementary Table 10. TRIPOD+AI reporting checklist mapping for the manuscript

Frontiers in Signal Processing R1 supplementary table

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Standalone display version for LaTeX upload/review. Machine-readable source: source_tables/Supplementary_Table_10_TRIPOD_AI_checklist.csv. Rows: 52; columns: 5. The display below contains the complete source table.

Table 1: Supplementary Table 10

item	topic	status	reported location	comment
1	Title	Reported	Title	Identifies self-supervised CTG trajectory representation and calibrated neonatal risk enrichment in the target intrapartum CTG population.
2	Abstract	Reported	Structured abstract	Structured Background Objectives Methods Results Conclusions abstract aligned with Frontiers in Signal Processing positioning.
3a	Healthcare context and rationale	Reported	Introduction	Explains intrapartum CTG decision-support context and limitations of static or subjective interpretation.
3b	Target population and intended purpose	Reported	Introduction and Discussion	Target population is intrapartum CTG records; intended use is risk enrichment and phenotype interpretation rather than autonomous diagnosis.
3c	Known health inequalities	Partially reported	Discussion limitations	No protected sociodemographic fairness analysis was possible beyond available CTU-UHB metadata; this is treated as a limitation.
4	Objectives	Reported	Introduction	Objective is evaluation of an open-data self-supervised CTG trajectory-representation workflow for calibrated neonatal risk enrichment.
5a	Data sources	Reported	Methods and Supplementary Table 1	CTU-UHB/CTU-CHB is primary raw-signal cohort; CTGDL-FHRMA is external representation/domain-shift resource; UCI CTG is feature-level reference.
5b	Data dates	Partially reported	Supplementary Table 1	Public datasets are cited with access information; detailed original accrual dates depend on source metadata.
6a	Study setting	Reported	Methods and Supplementary Table 1	Intrapartum cardiotocography public data from hospital-based CTG resources.
6b	Eligibility criteria	Reported	Methods and Supplementary Table 1	Records included when raw FHR/UC and required metadata were available for the analysis pipeline.
6c	Treatments received	Not applicable	Methods	No treatment-effect modelling was performed; delivery process variables were not used as intervention exposures.

Continued from Supplementary Table 10.

item	topic	status	reported location	comment
7	Data preparation and quality checking	Reported	Methods	Signal windowing, missingness, feature extraction, train-only scaling and source-data generation are described.
8a	Outcome definition and time horizon	Reported	Methods outcome definition	Primary endpoint is cord pH ¡7.15 or 5-minute Apgar ¡7 at record level.
8b	Subjective outcome assessment	Not applicable	Methods outcome definition	Outcome uses recorded pH and Apgar metadata from the public dataset rather than new subjective adjudication.
8c	Outcome blinding	Not applicable	Methods	Retrospective public-data analysis; outcome labels were withheld from SSL pretraining.
9a	Initial predictor choice	Reported	Methods outcome and prediction models	Classical signal features and SSL embeddings were prespecified from CTG physiology and self-supervised representation design.
9b	Predictor definitions	Reported	Methods and supplementary tables	Signal features, SSL embedding dimensions and phenotype variables are described; source-data tables are included.
9c	Subjective predictor assessment	Not applicable	Methods	Predictors were algorithmically extracted from public CTG signals and metadata.
10	Sample size	Reported	Methods and Results	Record counts, event counts and window counts are reported; limitations discuss small event count and high-dimensional model risk.
11	Missing data	Reported	Methods and Results	Median imputation, missingness features and no-missingness sensitivity are reported.
12a	Data partitioning and analysis use	Reported	Methods record split	Record-level training/test split, train-only SSL pretraining and held-out validation are stated.
12b	Predictor handling	Reported	Methods	Numeric predictors were median-imputed and standardized; categorical phenotypes were one-hot encoded.
12c	Model type and tuning	Reported	Methods and Supplementary Table 9	Transformer SSL, logistic pipelines, exploratory compact sweep and development-only selection sensitivity are described.
12d	Heterogeneity across clusters	Partially reported	Results and Discussion	Domain-shift analysis compares CTU-UHB with CTGDL-FHRMA; formal multi-centre heterogeneity was not possible.

Continued from Supplementary Table 10.

item	topic	status	reported location	comment
12e	Performance measures	Reported	Methods and Results	AUROC, AUPRC, Brier score, calibration, decision curve and bootstrap CIs are reported with rationale.
12f	Model updating or recalibration	Reported	Methods and Results	Cross-fitted Platt and isotonic recalibration are described; Platt scaling is emphasized.
12g	Prediction calculation	Reported	Methods and Code availability	Pipeline scripts are named and the versioned repository archive is available at https://doi.org/10.5281/zenodo.20485068.
13	Class imbalance	Reported	Methods and Results	Class-balanced logistic regression and AUPRC emphasis are justified under event imbalance.
14	Fairness	Partially reported	Discussion limitations	Available public metadata did not support a full fairness assessment; age and clinical stratification were limited.
15	Model output	Reported	Methods and Discussion	Outputs are risk-enrichment scores and calibrated probabilities; not intended as definitive diagnosis.
16	Training versus evaluation differences	Reported	Methods and Results	CTU train/test split and CTGDL representation-domain differences are explicitly described.
17	Ethical approval	Reported	Declarations	Public de-identified datasets; ethics approval not applicable.
18a	Funding	Reported	Declarations	No specific funding.
18b	Conflicts of interest	Reported	Declarations	No competing interests declared.
18c	Protocol	Reported	Code availability and limitations	No formal protocol was registered; analysis scripts and source data are included for reproducibility.
18d	Registration	Reported	Code availability and limitations	Study was not registered.
18e	Data sharing	Reported	Availability of data and materials	Public data sources and derived source-data tables are listed.
18f	Code sharing	Reported	Code availability	Local scripts are identified; the GitHub repository and versioned Zenodo archive are reported in Code availability.
19	Patient and public involvement	Not applicable	Methods/Declarations	No patient or public involvement in this retrospective public-data analysis.
20a	Participant flow	Reported	Results and Figure 1	Cohort scale and train/test records are shown.
20b	Participant characteristics	Reported	Supplementary Table 1	Dataset scale and available record-level characteristics are summarized in supplementary files.

Continued from Supplementary Table 10.

item	topic	status	reported location	comment
20c	Development versus evaluation comparison	Reported	Figure 1 and Supplementary Table 1	Train/test event balance and external domain-shift analysis are reported.
21	Number of participants and events per analysis	Reported	Results and Supplementary Table 9	Events are reported for primary and sensitivity analyses.
22	Model specification	Reported	Methods and Code availability	Architecture, feature dimensions and logistic models are described; implementation scripts are included in the GitHub repository and versioned Zenodo archive.
23a	Model performance with uncertainty	Reported	Results and Supplementary Tables 2-3	Performance estimates include bootstrap confidence intervals and model differences.
23b	Heterogeneity in performance	Partially reported	Results and Discussion	Domain-shift and missingness sensitivity are reported; subgroup fairness analysis remains limited.
24	Model updating results	Reported	Results and Supplementary Table 7	Platt and isotonic recalibration results are reported.
25	Interpretation	Reported	Discussion	Interprets risk enrichment, phenotype physiology, calibration and external-domain boundaries.
26	Limitations	Reported	Discussion	Discusses retrospective design, model-selection sensitivity, event count, dimensionality, missingness and lack of external outcome validation.
27a	Handling poor quality or unavailable inputs	Partially reported	Methods and Discussion	Missingness sensitivity and signal-quality features are reported; deployment handling is future work.
27b	User interaction and expertise	Reported	Discussion	Model is positioned for clinician-facing decision support rather than autonomous use.
27c	Future research	Reported	Discussion	Calls for larger locked train/development/test designs, external outcome validation and prospective evaluation.